DATA SCIENCE & MACHINE LEARNING · PROJECT 2

Project 2: Telco Customers

Predicting customer churn so the retention team calls the right people, before they go.

Jovana Krstevska · October 2026

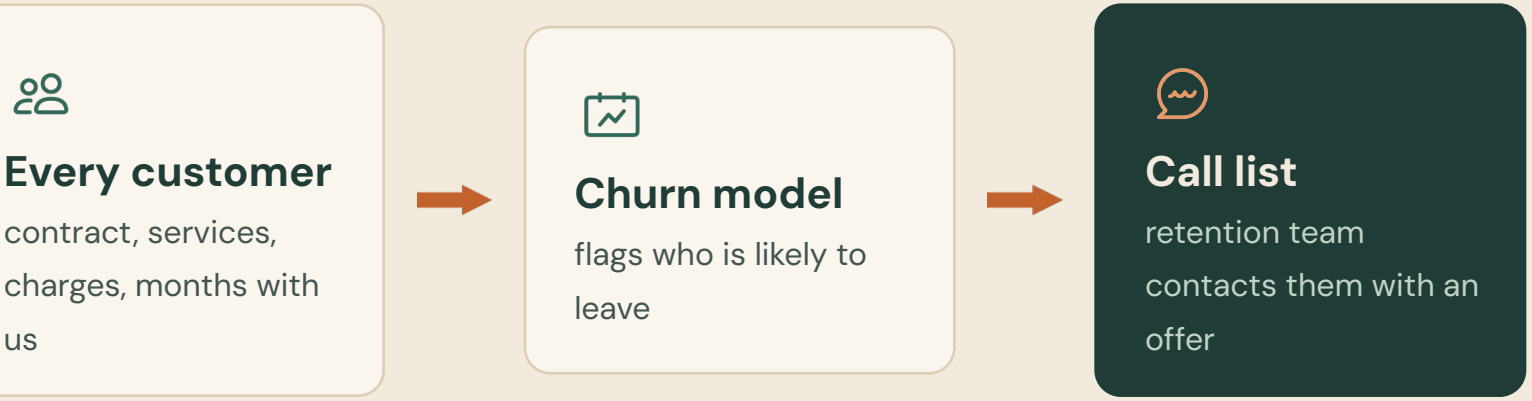


THE PROBLEM

Calling every customer is not feasible, so can we predict who is at risk of leaving?

26.5%

of 7,043 customers churned



The question: which current customers are likely to leave, so the team can reach them before they go?

THE DATA

One row is one customer, what else is hiding:

7,043
customers

19
features: contract, services, charges, tenure

80 / 20
stratified train / test split, fixed seed

1 Numbers stored as text

TotalCharges had 11 blanks, brand-new customers (tenure 0), so they were filled with 0.

2 A third answer

“No internet service” kept as its own category, so “No” means “has internet, but not this add-on”.

3 An ID is not a signal

customerID dropped. Everything else is known at the moment making a prediction.

Data: Telco Customer Churn (IBM sample), Apache 2.0 licence · 03

THE METRIC AND THE BAR TO BEAT

Accuracy is not the best metric here, so we score with F2

BASELINE STRATEGY 1

Say “everyone stays”

73.5% accurate

0 churners caught

Looks good, but is useless. F2 = 0.

BASELINE STRATEGY 2 · THE BAR TO BEAT

Call everyone

100% churners caught

0.644 F2 score

But 3 of every 4 calls go to someone who would have stayed.

Why F2: losing a customer costs more than an unnecessary offer, so F2 rewards catching churners more than avoiding wasted calls, but still punishes calling everyone.

MODELS TRIED

Six model families tried and the simplest tied for first

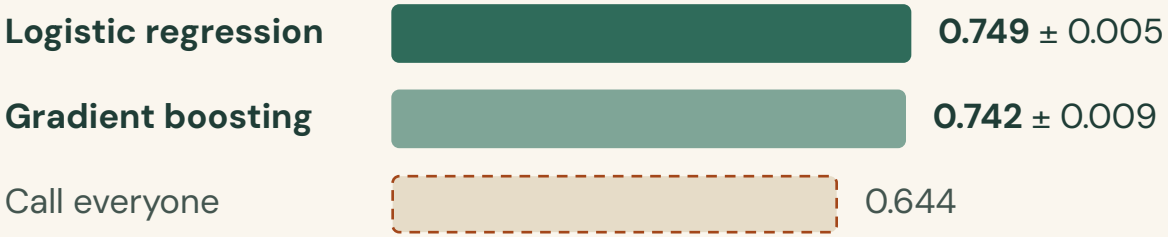
Compared on the same 5 folds

- Logistic regression
- k-nearest neighbours
- SVM
- Decision tree
- Random forest
- Gradient boosting

All six beat the baseline. The top two were tuned with a grid search on the training set.

> Built three new features. None improved the score by more than the noise between folds, so they were left out.

The final two, each at its own tuned threshold



F2 from nested cross-validation, scale 0 to 1

The gap of **0.007** is smaller than the noise between folds, so it's a tie. Choosing **logistic regression**: more stable, faster, and it is easier to work with and explain

THE DECISION THRESHOLD

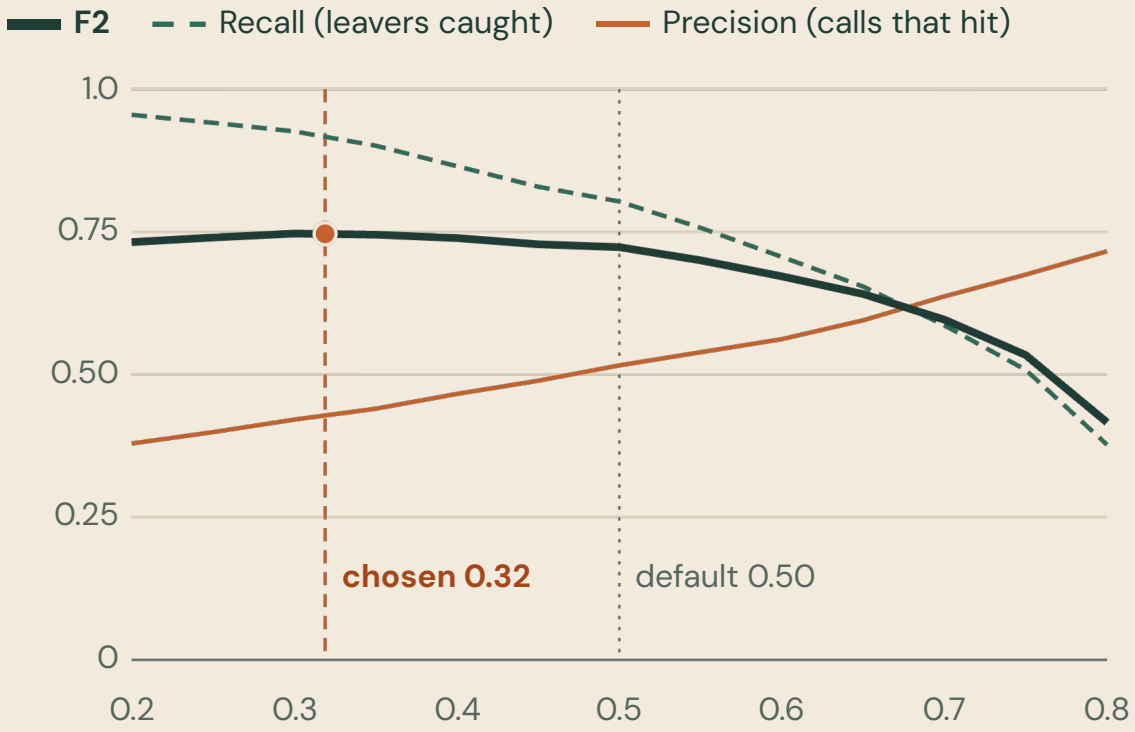
Lowering the bar from 0.50 to 0.32 to catch more churners

The model gives every customer a risk score.
The threshold decides who goes on the call list.

At 0.50 (the default)
80% of leavers caught, 294 missed

At 0.32 (the chosen threshold)
About 92% caught. Roughly 2 in 5 calls reach a real leaver.

Chosen to maximise F2, inside cross-validation, on training data.



Cross-validated predictions on the training set; x-axis is the threshold · 06

THE TEST, RUN ONCE

On 1,409 unseen customers, it caught 344 of 374 leavers

	Flagged · 779	Not flagged · 630
Actually left 374	**344** churners caught	**30** churners missed
Actually stayed 1,035	**435** calls not needed	**600** happy customers

0.756
F2 on the test set

0.749 ± 0.005
predicted by cross-validation

0.644
call-everyone baseline

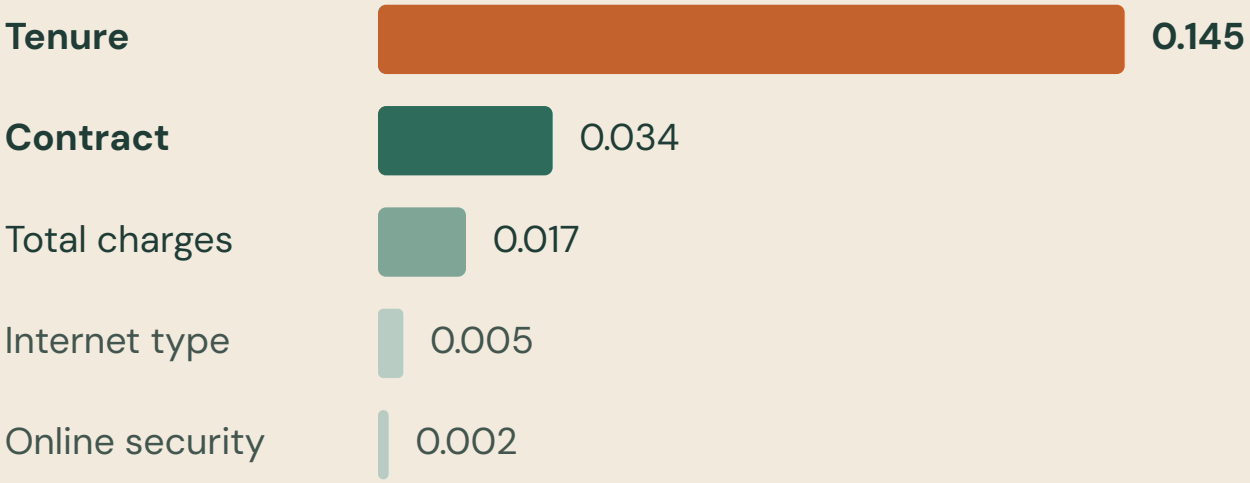
779 calls find 344 leavers. Calling at random would take about 1,300.

Final model: logistic regression, threshold 0.32 · test set used once · 07

WHAT DRIVES THE PREDICTIONS

How long someone has been a customer drives almost everything

Drop in test F2 when one feature is scrambled



Without tenure, the model falls below the baseline. Payment method and tech support mattered in the raw data, but their signal overlaps with tenure and contract.

The first months are riskiest

Onboarding and an early check-in are the biggest retention opportunity.

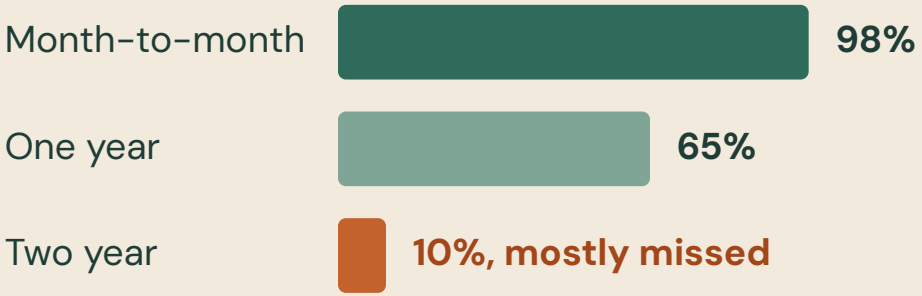
Monthly contracts churn the most

Offers that move people onto 1- or 2-year contracts are the most direct lever.

WHERE NOT TO TRUST IT

It misses the churners who look loyal

Share of churners caught, by contract



56 months, missed leavers

8 months, caught leavers

Flags 55% of customers
Too many to call? Prioritize from the highest risk and work down.

Scores are a ranking
0.6 doesn't mean a 60% chance. Use to order the list.

Who, not why
It can't tell which offer will make someone stay.

One snapshot
No dates in the data. Recheck and retrain on newer customers.

RECOMMENDATIONS

Used as a priority list, starting with new customers on monthly contracts

Recommendations:

1. Call from the top of the risk ranking down, as far as the capacity allows.
2. Invest in the first months: onboarding and an early check-in.
3. Offer monthly customers a reason to move to a longer contract.

Next Steps:

- Set the threshold from real costs: the value of a kept customer vs the cost of an offer.
- Find signals for loyal-looking churners, eg: complaints or recent usage.
- Test it on newer customers.

Thank you!

(demo)

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